

		Section A	Marks	CO	BL
Q1	(a)	Acharya P.C. Ray founded which company? a) Cipla b) Bengal Chemicals & Pharmaceuticals c) Sun Pharma d) CSIR Labs	1	1	1
	(b)	C.N.R. Rao is known for major work in: a) Genetics b) Nanomaterials and solid-state chemistry c) Astronomy d) Polymer engineering	1	1	1
	(c)	Har Gobind Khorana received Nobel Prize for work in: a) Ribosome structure b) Genetic code interpretation c) Inorganic chemistry d) Magneto-chemistry	1	1	1
	(d)	Shanti Swarup Bhatnagar is known as: a) Father of Indian Nanotech b) Father of Research Laboratories in India c) Father of Biotechnology d) Father of Physics	1	1	1
	(e)	R.A. Mashelkar pioneered work in: a) Protein folding b) Non-Newtonian fluids and polymer science c) Coal chemistry d) Soil analysis	1	1	1
Q2	(a)	Calculate the carbonate and non-carbonate hardness of a sample of water in ppm containing: Ca(HCO ₃) ₂ = 16.2 mg/L, Mg(HCO ₃) ₂ = 14.6 mg/L, MgSO ₄ = 48 mg/L, MgCl ₂ = 95 mg/L, KCl = 100 mg/L, CO ₂ = 22 mg/L	5	2	3
	(b)	Calculate the quantities of Lime & Soda (90% pure each) in kilogram, required for treatment of 25,000 litres of hard water, containing following Ca ⁺⁺ = 30 mg/L, Mg ⁺⁺ = 21.6 mg/L, H ₂ SO ₄ = 4.9 mg/L, CO ₂ = 4.4 mg/L, HCO ₃ ⁻ = 12.2 mg/L, Fe ₂ O ₃ = 15.4 mg/L.	10	2	3
Q3	(a)	Giving structure of EBT, EDTA-M ⁺⁺ complex and reactions, explain how can the hardness of water be determine by EDTA method.	5	2	2
		OR			
		Write detailed note on following: (Give reactions too) (i) Caustic embrittlement (ii) Calgon and phosphate conditioning	5	2	2
	(b)	Describe Lime-Soda method of softening of water under following head: (i) Principle (ii) Chemical reactions (iii) Diagram (iv) Advantages and disadvantage	10	2	2
		OR			
		Describe Ion Exchange method of softening of water under following head: (i) Principle (ii) Chemical reactions (iii) Labelled diagram (iv) Regeneration of resin	10	2	2
Section B					
Q4	(a)	A good lubricant should have: a) Low viscosity index b) High viscosity index c) High pour point d) Low flash point	1	3	1
	(b)	The best fuel should have: a) High ignition temperature b) Low calorific value c) High calorific value d) None	1	4	1
	(c)	Gross calorific value (GCV) is always: a) Less than LCV b) Greater than LCV c) Equal to LCV d) Unrelated to hydrogen content	1	4	2
	(d)	Which coal has highest carbon and calorific value? a) Peat b) Lignite c) Bituminous d) Anthracite	1	4	2
	(e)	If a fuel has high hydrogen content, its GCV will: a) Increase b) Decrease c) Not changed d) Become zero	1	4	2
			P.T.O		

Time: 3 hrs

Q.5	(a)	A lubricating oil of unknown viscosity-index has the Saybolt Universal Viscosity of 58 seconds at 210°F and of 564 seconds at 100°F. The standard pennsylvanian oil has Saybolt Universal Viscosity of 58 seconds at 210°F and 420 seconds at 100°F. The standard gulf oil has a Saybolt Universal Viscosity of 58 seconds at 210°F and 758 seconds at 100°F. Calculate the viscosity-index of oil.	5	3	3
	(b)	Calculate the gross and net calorific values of coal having the following composition: Carbon: 70%, Hydrogen: 8%, Sulphur: 1%, Nitrogen: 0.5%, Oxygen: 20%, Ash :0.5%.	10	4	3
Q.6	(a)	Define Calorific value? Giving formula of calculation and description of each constituent, discuss in detail the Proximate analysis of coal.	5	4	2
		OR			
		Describe following properties of lubricants with their significance: (i) Flash and Fire Point (ii) Cloud and Pour point	5	3	2
	(b)	Write any 4 functions of lubricants. Discuss mechanism of lubrication by explaining: (i) Fluid film lubrication (ii) Boundary lubrication	10	3	2
		OR			
		Giving chemical structure of any 2 solid lubricants, discuss the chemistry how they are serving as lubricants.	10	3	2

Blooms Texonomy Level CH10010



■ BL1 ■ BL2 ■ BL3